Sunny (Zehui) Xu
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RESEARCH INTEREST

Distributed control and coordination of multi-agent robotic systems; safety-critical control and collision avoidance; integrating learning-based methods with model-based control for safe autonomy; simulation and hardware validation of collective robotic behavior.

EDUCATION

University of Michigan — Ann Arbor, MI
B.S.E. in Computer Science and B.S.E. in Robotics — *Expected: December 2026*
GPA: 3.93 / 4.00
Coursework: Data Structures and Algorithms, SLAM and Navigation, Robot Operating Systems, Sensors and Signals, Deep Learning for Robot Perception, AI and Ethics

PUBLICATION

Zehui Xu*, Xinyue Xu*, and Steven Ceron. "3D Robotic Swarmalators that Reconfigure, Navigate, and Avoid Obstacles"
IEEE International Conference on Robotics and Automation (ICRA), 2026. (Published)

RESEARCH EXPERIENCE

Undergraduate Research Assistant — Ann Arbor, MI
Synergetic Adaptive Machinas (SAMS Lab) — *May 2025 – present*
Advisor: Professor Steven Ceron

3D Robotic Swarmalators — ICRA 2026

- Co-first-authored and presented a paper on distributed control of 3D robotic swarmalators at ICRA 2026.
- Implemented a 3D swarmalator controller with control barrier functions (CBFs) for distributed drone coordination and collision avoidance in obstacle-rich environments.
- Integrated Rodrigues' rotation formula into the swarmalator controller to orient the drone collective, allowing drones to pass through obstacles more easily.
- Developed real-time control software integrating the Crazyflie API, ROS communication, and VICON motion capture feedback for closed-loop control of 8 drones.

Smarticle Collective Simulation

- Built an NVIDIA Isaac Sim environment using SOLIDWORKS-derived URDF models to study stochastic collisions and collective behavior of 100+ smarticle robots.
- Automated multiple headless simulation instances using Bash scripts, enabling unattended overnight experiments.
- Developed a randomized initialization script that accounted for smarticle geometry and range of motion to place robots within a bounded region without initial overlap.
- Documented simulation setup and workflows to enable reuse and adaptation for future lab projects.

GPU-Accelerated Particle-Life Simulation

- Scaled a particle-life simulation from an initial capacity of 3,000 to 500,000+ particles, simulating 45 seconds of particle dynamics in under 10 minutes.
- Implemented GPU-parallel physics computations and spatial partitioning to accelerate particle interaction calculations and controller updates.
- Redesigned particle interaction calculations to target O(n) scaling by restricting computations to spatially local neighbors.

EEG-Controlled Microrobot Collective (Capstone)

- Evaluated and selected EEG hardware for a brain-computer interface to control magnetic microrobot collectives.
- Integrated a controller with the microrobot physics simulation to translate directional commands into collective movement, supporting future integration with EEG input and the physical magnetic-array platform.
- Developing a signal-processing and classification pipeline combining Butterworth bandpass filtering with a two-stage machine learning model to map raw EEG signals to directional commands.

TEACHING EXPERIENCE

Deep Learning for Robot Perception Teaching Assistant Ann Arbor, MI
College of Engineering *Jan 2025 – August 2025*

- Held office hours to explain deep learning concepts and support PyTorch projects in image classification, object detection, and pose estimation.
- Led a discussion session on simulation and reinforcement learning for robotics.
- Graded examinations in collaboration with course instructors.

General Physics II Teaching Assistant Ann Arbor, MI
Literature, Science, and Arts *Jan 2025 – December 2025*

- Guided students through electromagnetism concepts and problem-solving, including electric and magnetic fields, circuit analysis, and electromagnetic waves.
- Adapted explanations to different problem-solving approaches, helping students identify conceptual gaps and refine their reasoning.

EXPERIENCE

Robotics Software Intern Burlingame, CA
Neptune Medical *May 2026 – August 2026*

- Redesigned a Python-based fault audit system using clinical feedback, presenting faults in clear pass/fail summaries and reducing clinical documentation time for faulty runs from 60 to 10 minutes.
- Developed a Python log-analysis tool for an FDA-regulated endoscopy robot to detect overlooked failures and performance degradation to inform R&D researchers and service technicians.
- Built data pipelines for log ingestion, storage, and scheduled analysis to detect short- and long-term device failures, preparing the monitoring system for deployment across multiple locations.
- Presented the debugging tool at a company-wide meeting, receiving positive feedback from R&D and leadership.

Robotics Software Subteam Lead
Wolverine Botkickers *January 2026 - Present*

- Developed and validated an iterative landmark-matching algorithm in Gazebo for humanoid pose estimation from noisy soccer-field observations.
- Led the redesign of an A* planner with a custom ROS2 interface; validated functionality and compared search performance against BFS through simulation and automated Bash tests.
- Developed a ROS2 point-cloud publisher to simulate continuous lane-line observations for computer vision integration.

Autonomy Developer
MBot SLAM & Navigation Project *September 2024 – December 2024*

- Integrated SLAM and path planning on a Raspberry Pi using LCM, enabling real-time mapping and autonomous navigation.
- Implemented particle-filter localization using lidar measurements and odometry, alongside occupancy-grid mapping and PID control for autonomous exploration.

HONORS AND AWARDS

CRA Outstanding Computing Research Undergrad Award - Honorable Mention	2026
James B. Angell Scholar	2026
University Honors	2023-2026
Dean's List	2023-2025
William J. Branstrom Freshman Prize	2024

SKILLS

- *Programming languages:* C++, C, Python, Java, JavaScript, MATLAB
- *Operating systems:* Windows, Linux, macOS
- *Tools:* Onshape, SOLIDWORKS, EasyEDA, Git/GitHub
- *Robotics & Software:* ROS2, NVIDIA Isaac Sim, Docker, PyTorch, OpenCV, Gazebo, React, HTML, SQL, LaTeX